NUMERICAL ANALYSIS
GUIDED LAB 1

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1. Compute the absolute and relative error for

$$\sqrt{2} \approx 1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3}$$

Solution. Let's first compute the left and right hand side of the equation above using x_{approx} to represent the approximate solution (RHS). Using matlab we see that

$$\begin{aligned} x &\approx 1.41421356237, \text{ and} \\ x_{\text{approx}} &= 1.4142129630 \end{aligned}$$

Let ε_{abs} represent the absolute error, and ε_{rel} represent the relative error. Then we have the following:

$$\begin{aligned} \varepsilon_{\text{abs}} &= |\sqrt{2} - x_{\text{approx}}| \\ &= |1.41421356237 - 1.4142129630| = 5.99 \times 10^{-7} \end{aligned}$$

and we also see that:

$$\begin{aligned} \varepsilon_{\text{rel}} &= \frac{\varepsilon_{abs}}{\sqrt{2}} \\ &\approx \frac{5.99 \times 10^{-7}}{1.41421356237} \\ &\approx 4.24 \times 10_{-7} \end{aligned}$$

□

2. The Newton-Raphason method for determining $\sqrt{a}$ follows from repeated iteration of

$$y_{n+1} = \frac{1}{2}\left(y_n + \frac{a}{y_n}\right)$$

Show that $y = \sqrt{a}$ is the fixed point of this sequence.

Proof. To achieve $y = \sqrt{a}$ as a fixed point of our equation we will need to compare the error between the n^{th} term and the $(n+1)^{\text{st}}$ term.

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We begin by subtracting $\sqrt{a}$ from both sides of the equation and performing some algebraic manipulation. Consider

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) - \sqrt{a} \\ &= \frac{x_n^2 - 2x_n\sqrt{a} + a}{2x_n} \\ &= \frac{(x_n - \sqrt{a})^2}{2x_n} \end{aligned}$$

Accordingly,

$$\begin{aligned} \lim_{n\to\infty} \frac{|x_{n+1} - \sqrt{a}|}{|x_n - \sqrt{a}} &= \lim_{n\to\infty} \frac{1}{2x_n} \\ &= \frac{1}{2\sqrt{a}} \end{aligned}$$

Hence the sequence generated by this scheme has order of convergence equal to 2 and asymptotic error constant $1/(2\sqrt{a})$. □

3. Create a script that implements Heron's algorithm (the formula in problem 1). It should take as the input: (1) the number a to square root, (2) the initial guess y_0, and (3) the total number N of iterations. In addition to outputting the approximate square root at each iteration. you may use a bulit-in functionas the "exact" answer for comparison. Plot bothe the relative error as a function of iteration for $N = 6$ on a log-log scale for $a = 2$. How sensitive is the result to y our initial guess of y_0?

Solution. Consider a function that we have in file "problem3.m" that we have below:

```
function [p,absError,relError] = fixpt(a,y,n)
    % Input - a, the number to sqaure root
    %       - y, the initial guess
    %       - n, the number of iterations to run
        p = y; % Initialize p with the initial guess
        for k = 1:n
            p_new = 0.5 * (p + a / p);
            absError = abs(p_new - p);
            relError = absError / abs(p_new);
            if absError < 1e-10
                break;
            end
            p = p_new;
        end
```

```
end
```

So we see our plot from the above results as: But, if we fiddle with the initial value for the

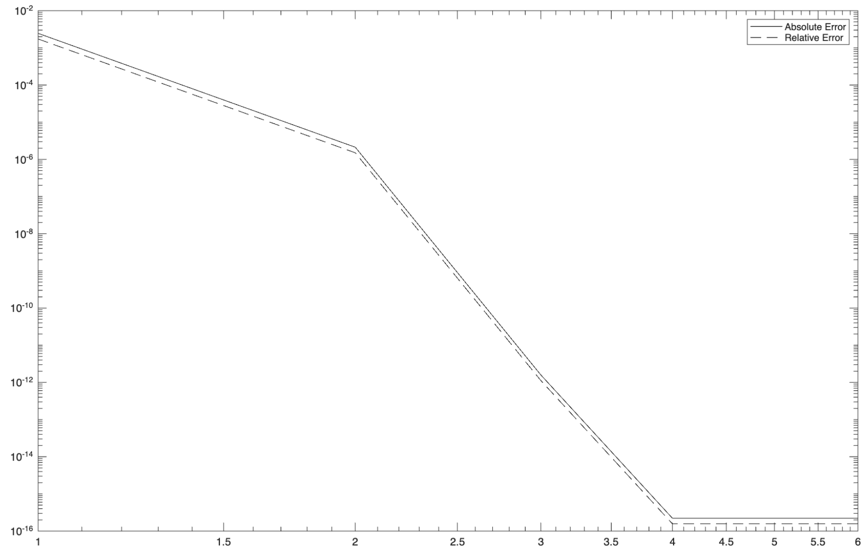

FIGURE 1. Log-Log plot of the absolute and relative error for Newton Raphson for $N = 6$, $y_0 = 1.5$

answer we see different errors. Consider Figure 2 which shows the errors when we set $y_0 = 30$

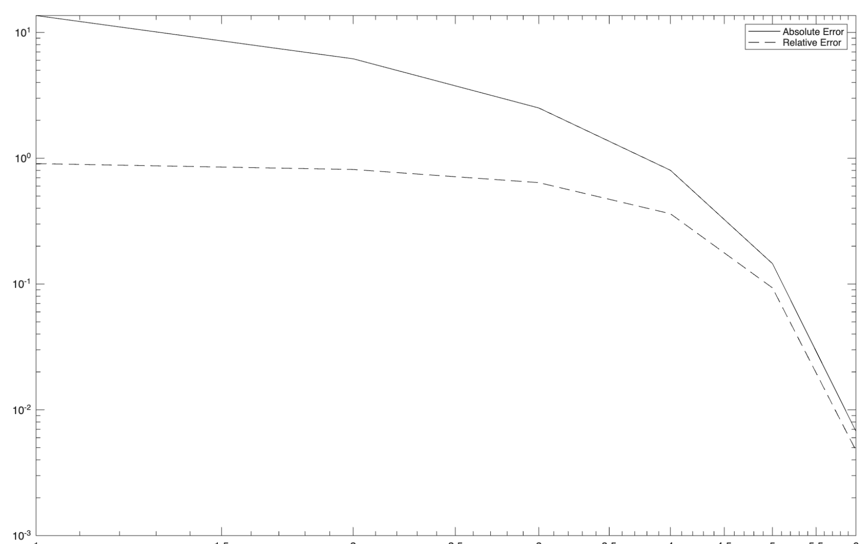

FIGURE 2. Log-Log plot of the absolute and relative error for Newton Raphson for $N = 6$, $y_0 = 30$

and we are done. $\square$

4.

Solution. Problem 4 □

5.

Solution. Problem 5 □

6.

Solution. Problem 6 □

7.

Solution. Problem 7 □

8.

Solution. Problem 8 □

9.

Solution. Problem 9 □

10.

Solution. Problem 10 □